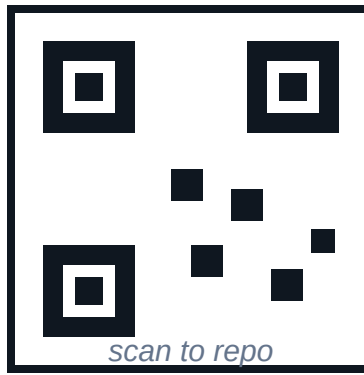




A Unified Control Plane for Machine-Learning Security Services

An Authenticated Gateway Fronting Heterogeneous ML Security Products

Pooja Kiran

Security Engineering • AI / Agent Security • Cloud Security

Apache-2.0 • Python 3.12 • HEAD 7ac348c • verified 2026-09-26

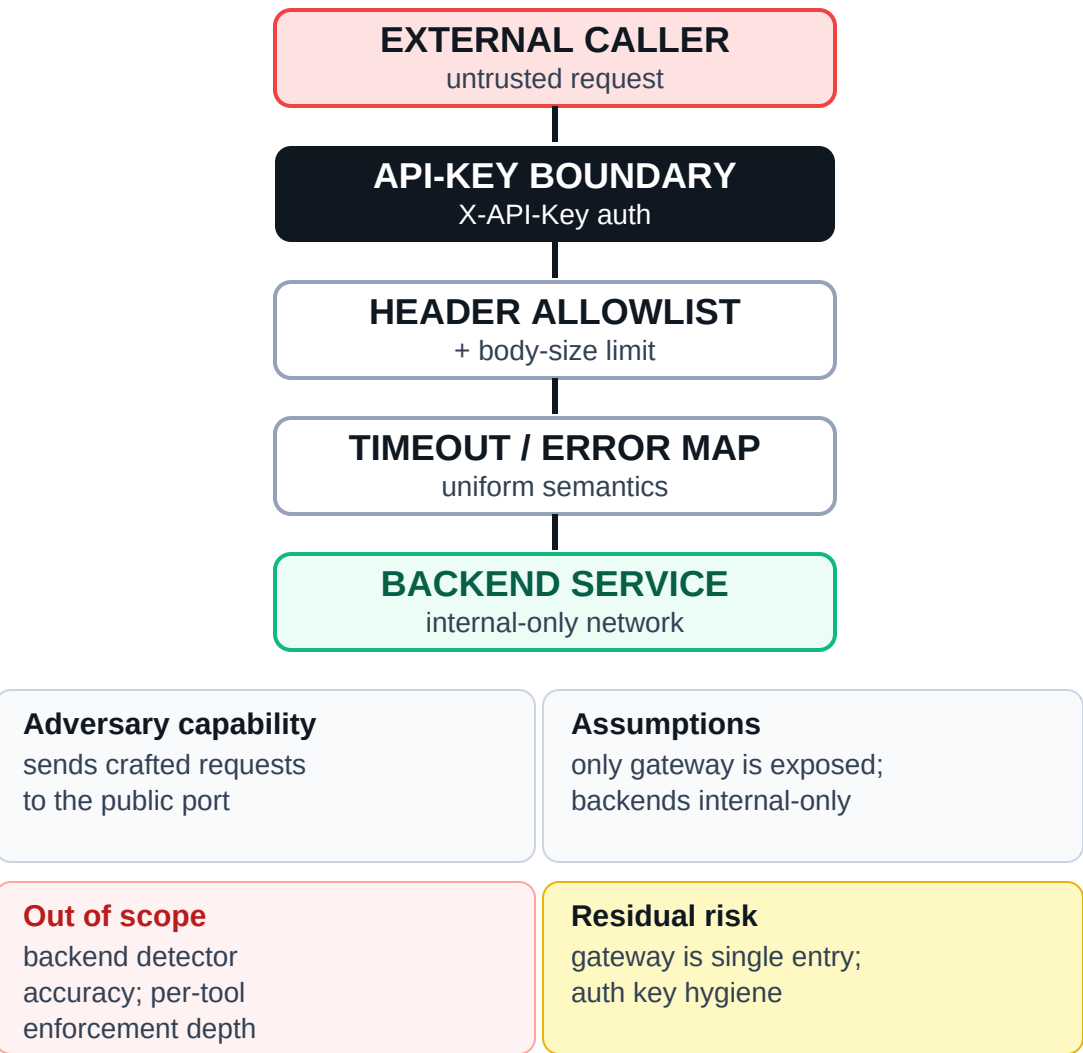
RESEARCH QUESTION

Can heterogeneous ML security services be fronted by one authenticated control plane with consistent auth, limits, and error handling?

01 • THE FAILURE MODE

Each ML security tool ships its own HTTP service, auth model, and error semantics. Wiring them into a product means every caller re-implements auth, header handling, body limits, and timeouts — inconsistently. A single gateway normalizes the entry boundary for the three real backend services.

02 • THREAT MODEL



03 • OBJECTIVE

Determine whether one FastAPI gateway can provide consistent auth, allowlisting, limits, and error mapping across three ML security backends.

- 01 – X-API-Key caller auth
- 02 – Header allowlist + body-size limit
- 03 – Timeout + uniform error mapping
- 04 – Internal-only backend network

HERO • SYNCHRONOUS CONTROL PLANE

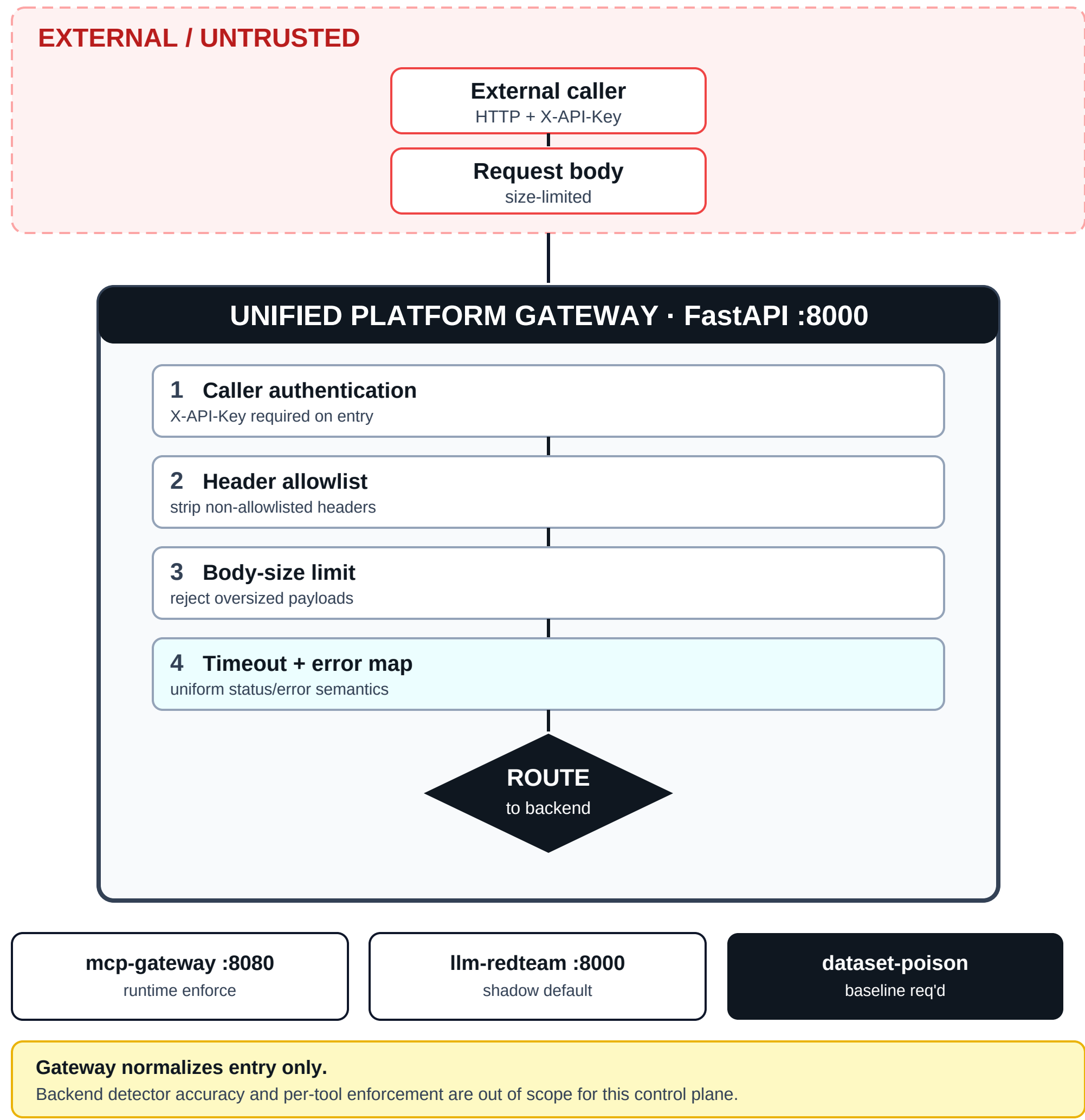
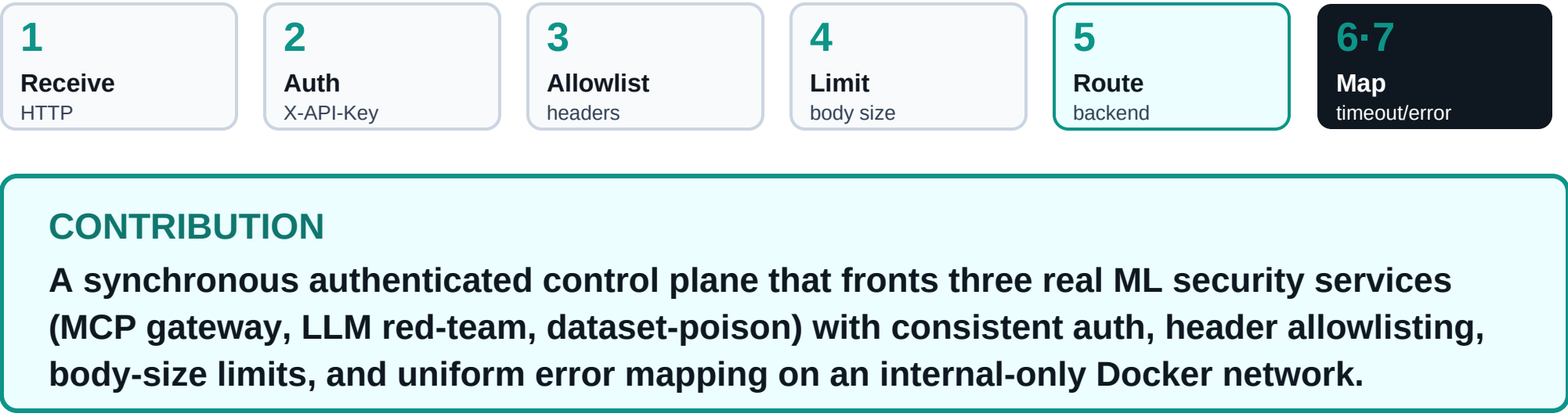


Figure 1. Synchronous control plane: one authenticated FastAPI gateway fronts three real backends on an internal-only network.

METHOD • REQUEST HANDLING



RESULTS • CURRENT LOCAL VALIDATION

59 TEST FUNCTIONS

3 REAL BACKENDS

59 test_ functions across the platform test suite. Three integrated HTTP products behind one gateway.

Verified surface

Public host ports 1

Auth mechanism API key

Compose files 2

ENTRY-BOUNDARY CONTROLS

Caller auth 100%

Header allowlist 100%

Body-size limit 100%

Error mapping 100%

Controls implemented at the gateway entry. Backend enforcement depth is per-tool, not sho

ARCHITECTURE PROVENANCE

ARCHITECTURE.md documents a synchronous control plane + separate batch/admission jobs. Production Docker network is internal-only; gateway alone exposed.

NOT ESTABLISHED BY THIS REPO

Backend detector accuracy. Distributed orchestration. Production reliability, uptime, or throughput SLOs.

SECURITY BOUNDARY

PROTECTS / DETECTS

authenticated entry

header allowlist

body-size limits

uniform errors

backend isolation

DOES NOT ESTABLISH

backend accuracy

async orchestration

distributed scale

production SLA

per-tool enforce

ONE AUTHENTICATED ENTRY

LIMITATIONS

- 1. Gateway normalizes entry, not backend accuracy.
- 2. Synchronous control plane; not an async orchestrator.
- 3. Single public entry point (blast-radius consideration).
- 4. Local tests ≠ production reliability.
- 5. Per-tool enforcement depth varies (documented).

FUTURE WORK

- Async / queued orchestration mode.
- Per-backend rate limiting.
- Distributed multi-replica state strategy.
- End-to-end integration benchmark.
- mTLS between gateway and backends.

CONCLUSION

The platform provides one authenticated FastAPI gateway that normalizes auth, allowlisting, limits, and errors across three real ML security backends. It integrates services; it does not orchestrate them asynchronously or assert backend accuracy.

REPRODUCE

docker compose up -d

pytest tests/